

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2-DEBUGGING & OPTIMISATION TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO2	Assessment:	ASSESSMENT TOOL3- DEBUGGING & OPTIMISATION TASK
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems. (BL3)</i>		
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ANNEXURE A

1. A program performing signed integer addition using 2's complement produces incorrect results when adding large values (e.g., +120 and +50 in 8-bit representation). Debug the issue by analyzing the binary computation and identifying whether overflow has occurred. Propose a solution to correctly detect and handle overflow in such cases.
2. A processor using a ripple carry adder shows significant delay in executing arithmetic operations in a real-time system. Debug the performance bottleneck by examining how carry propagates through each bit. Suggest an optimized solution using a carry look-ahead adder, and explain how it reduces delay.
3. A multiplication unit implementing Booth's algorithm gives incorrect results for certain negative operands. Debug the step-by-step execution of the algorithm, focusing on bit-pair checking, arithmetic shifts, and sign extension. Identify the possible source of error and suggest corrections to ensure accurate signed multiplication.
4. An embedded system using the restoring division algorithm experiences inefficiency due to repeated restoration steps, leading to increased execution time. Debug the algorithm's workflow and identify where unnecessary operations occur. Propose an optimized approach using the non-restoring division method, explaining how it improves performance.
5. A scientific application using IEEE 754 single precision floating-point representation produces inaccurate results when adding very small and very large numbers. Debug the issue by analyzing exponent alignment, normalization, and rounding errors. Suggest optimization techniques, such as using double precision or algorithmic adjustments, to improve accuracy.

Code of Conduct:

I SARU HASINI.S (1 9 2 5 2 1 4 1 8) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SARU HASINI.S

MARKS ALLOCATION

	Identification of Errors / Bugs(20%)	Debugging Approach & Analysis(20%)	Correctness of Debugged Solution(25%)	Optimization Techniques Applied(20%)	Testing and Documentation(15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Identification of Errors / Bugs	20%	Accurately identifies all errors and issues with complete understanding of the problem	Identifies most errors with minor omissions	Identifies limited errors; misses key issues	Unable to identify errors or misunderstands the problem
Debugging Approach & Analysis	20%	Uses effective debugging techniques with clear analysis and root cause identification	Applies suitable debugging methods with minor gaps	Uses basic debugging methods with limited analysis	No systematic debugging approach followed
Correctness of Debugged Solution	25%	Provides completely corrected solution with accurate functionality and expected output	Provides working solution with minor errors	Partially correct solution with limited functionality	Incorrect solution or fails to resolve errors
Optimization Techniques Applied	20%	Applies efficient optimization methods to improve performance, memory usage and quality	Performs optimization with noticeable improvements	Limited optimization with minimal improvements	No optimization applied or inefficient implementation
Testing and Documentation	15%	Performs complete testing with proper validation and clear documentation	Provides adequate testing and documentation with minor issues	Limited testing and incomplete documentation	No proper testing or documentation provided

ANSWERS

1. Debugging 2's Complement Overflow

Debugging:

Add +120 and +50 in 8-bit.

Binary: $01111000 + 00110010 = 10101010$.

The result becomes negative, indicating overflow because the actual sum (170) exceeds +127.

Solution:

Detect overflow by checking if two positive numbers produce a negative result (or two negative numbers produce a positive result).

Display an overflow error or use a larger data type (16-bit/32-bit).

2. Debugging Ripple Carry Adder

Debugging:

Observe that each carry waits for the previous carry.

This creates a long propagation delay and slows arithmetic operations.

Solution:

Replace the Ripple Carry Adder with a Carry Look-Ahead Adder (CLA).

CLA generates carry signals in parallel, reducing delay and improving speed.

3. Debugging Booth's Algorithm

Debugging:

Verify bit-pair checking (00, 01, 10, 11).

Check arithmetic right shift and sign extension.

Incorrect handling of negative numbers causes wrong multiplication.

Solution:

Perform correct bit-pair operations.

Use arithmetic right shift with proper sign extension.

Validate results using signed operands.

4. Debugging Restoring Division

Debugging:

Identify repeated restoration after every negative subtraction.

These extra operations increase execution time.

Solution:

Use the Non-Restoring Division Algorithm.

It avoids unnecessary restoration, reducing the number of operations and improving performance.

5. Debugging IEEE 754 Floating-Point Addition

Debugging:

Check exponent alignment before addition.

Verify normalization and rounding after addition.

Precision is lost when adding very small and very large numbers.

Solution:

Use double precision (64-bit) instead of single precision.

Apply proper rounding and normalization to improve accuracy.